

Knowledge Distillation with Masked Autoencoders for Improved Model Privacy

Based on Yutong Bai, et al., "Masked Autoencoders Enable Efficient Knowledge Distillers"

Seminar Paper

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Abstract

Knowledge Distillation is the process of transferring knowledge from a larger model to a smaller model. Performance of the larger model is lost during the distillation process. The paper titled "Masked Autoencoders Enable Efficient Knowledge Distillers" explores the idea of Masked Autoencoders (MAE) showing better performance in knowledge distillation tasks compared to directly distilling knowledge from a large model.

We explore the idea of MAE based distilled models (DMAE) being more resilient to Membership Inference Attacks (MIA) compared to directly distilled models, models trained from scratch and MAE without Knowledge Distillation. We show that DMAE is inherently more resilient to MIA compared to other methods. With the teacher and the MAE model pretrained on CIFAR-100 and the Classifiers trained on CIFAR-10, The DMAE model, when stopped at minimum validation loss, shows a significant improvement in privacy with an Attacker's Prediction AUC of 0.5570, outperforming the directly distilled ViT model with an Attacker's Prediction AUC of 0.5989. Whereas this model reaches an average accuracy 73.47%, whereas ViT with feature distillation reaches an accuracy of 72.73%. But when trained further, the validation accuracy of DMAE increases to 76.45% but it also comes with a privacy trade-off where the Attacker's Prediction AUC reaches 0.6929. Implementation of this paper can be found at <https://github.com/I3th4I/DMAE-Keras>

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Abbreviations

DMAE	Distill Masked Auto Encoder
KD	Knowledge Distillation
MAE	Masked Autoencoder
MIA	Membership Inference Attack
MIM	Masked Image Modeling
MLM	Masked Language Modeling
NLP	Natural Language Processing
PPGF	Privacy-Preserving Generative Framework
VAE	Variational Autoencoder
ViT	Vision Transformer

1 Introduction

With the recent increase in the usage of online services and AI-based applications, organizations collect vast amounts of private/sensitive data related to individuals. This data is typically kept within the organization and is used to train AI models. While the data itself is not made public and is kept private, the trained models are often released to the public and used in various applications. Even though an attacker trying to get access to such data, may not be able to do so directly, the outputs produced by these models can potentially leak information that could help the attacker determine if a particular input sample/individual was part of the original data used for training the AI model. Such an attack is referred to as Membership Inference Attack (MIA).

Transformers, widely used for natural language processing, have also demonstrated significant performance in computer vision tasks. Masked Autoencoder (MAE), which operate on the principles of Masked Image Modeling, train models to predict the masked portions of an input image based on the unmasked regions. This approach enables the training of high-capacity vision transformer models with meaningful feature representations. MAEs have been shown to improve the performance of vision transformers in downstream classification tasks. Additionally, recent works have built upon this by applying knowledge distillation, one of the most popular model compression techniques to transfer knowledge from larger capacity Vision Transformer (ViT) to smaller ones. These Distill Masked Auto Encoder (DMAE)(Bai et al., 2023) have shown to further enhance accuracy of ViTs.

In this project, we show that DMAE enables us to create downstream classifiers that are less prone to MIA. This stems from the idea that DMAE creates robust latent representations of the input images, which are efficient in capturing the semantic information of the image as a whole and are invariant to pixel-level subtleties. Although, in theory, MAE should already help create downstream classifiers that are more resilient to MIA, our results show that it does not offer any significant improvement over vanilla ViT (without any pre-training). While previous studies show that susceptibility to MIA can be inherited from the teacher model through knowledge distillation, this issue can be easily avoided. In fact, we can even decrease susceptibility to MIA by training the teacher and student on different (mutually exclusive) datasets. We used CIFAR-100 for pre-training the teacher model and MAE, while we trained the downstream classifier on CIFAR-10. Both of the dataset are mutually exclusive.

2 Related Works

2.1 Knowledge Distillation (KD)

First proposed by , Buciluă et al., 2006, Knowledge Distillation (KD) is a popular model compression technique which allow us to have both the performance of a large model and also the fast inference times of small models. The KD framework includes a small student model, which is supervised by a large teacher model. KD methods can be classified into two branches:

2.1.1 Response-Based Knowledge Distillation

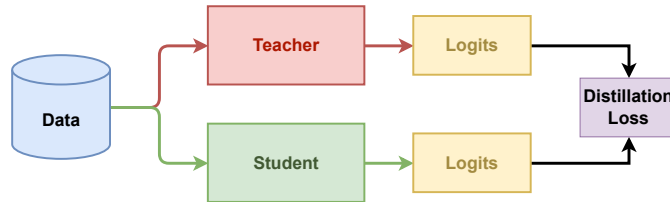


Figure 1 Illustration of the relationship between the teacher and student models in Response-Based Knowledge Distillation. The student model learns to mimic the output logits of the teacher model.

Here we directly take the final output logits of the teacher network and the student network triess to mimic this directly (Cho and Hariharan, 2019; Furlanello et al., 2018; Mirzadeh et al., 2020; Xiao et al., 2022; Zhang et al., 2018; Zhao et al., 2022).

2.1.2 Feature-Based Knowledge Distillation

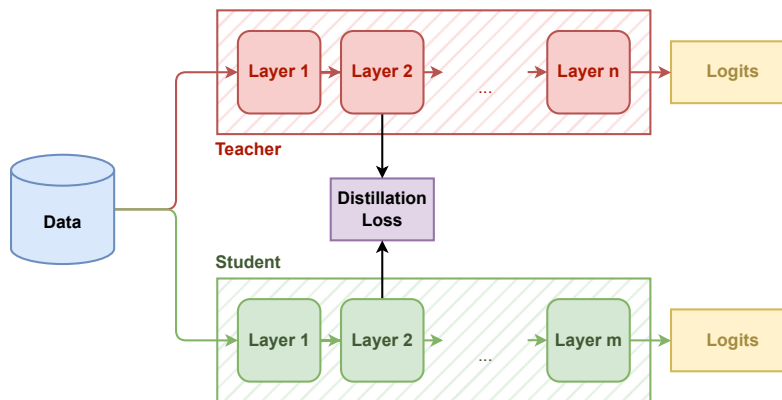


Figure 2 Illusstration of the Feature-Based Knowledge Distillation process. The student model learns by minimizing the difference between the intermediate layer activations of the teacher and student models.

This technique was first introduced in Fitnets (Romero et al., 2015), it works by directly minimizing the distance between the outputs of intermediate layer activations of the teacher model and the student model.

2.2 Masked Image Modeling (MIM)

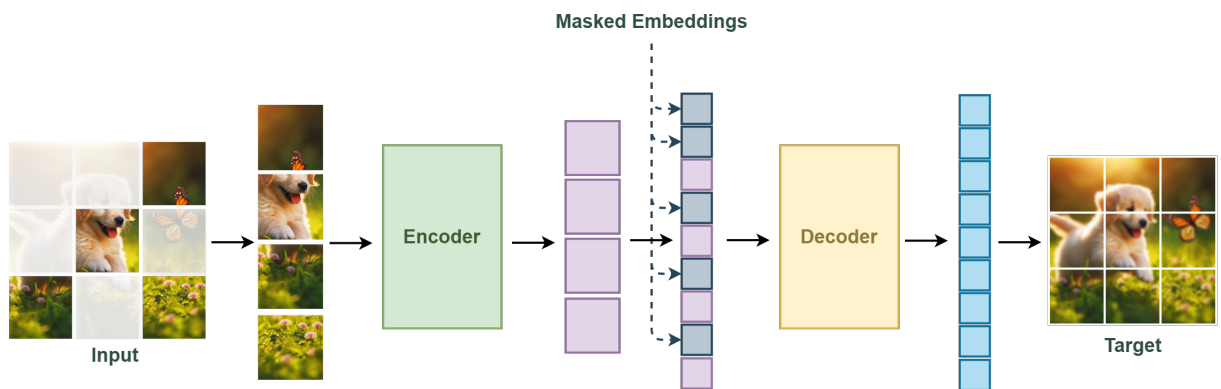


Figure 3 Illustration of the Masked Autoencoder (MAE) process. A portion of the input image is masked, and the model is trained to reconstruct the missing pixels.

"Masked Signal Modeling" refers to the technique of masking a part of the input signals, and then trying to predict these masked signals. With BERT (Devlin et al., 2019) showing promising improvements by extending this idea to NLP tasks, and ViTs being widely adapted for computer vision tasks, a set of techniques emerged in the area of using MIM for pre-training ViTs. BEiT (Bao et al., 2022) was the first technique to do this successfully, by learning to predict visual tokens. SimMIM (Xie et al., 2022) and MAE (He et al., 2022) later extend on this idea by directly predicting the pixels of the masked image patches.

2.3 Membership Inference Attack (MIA)

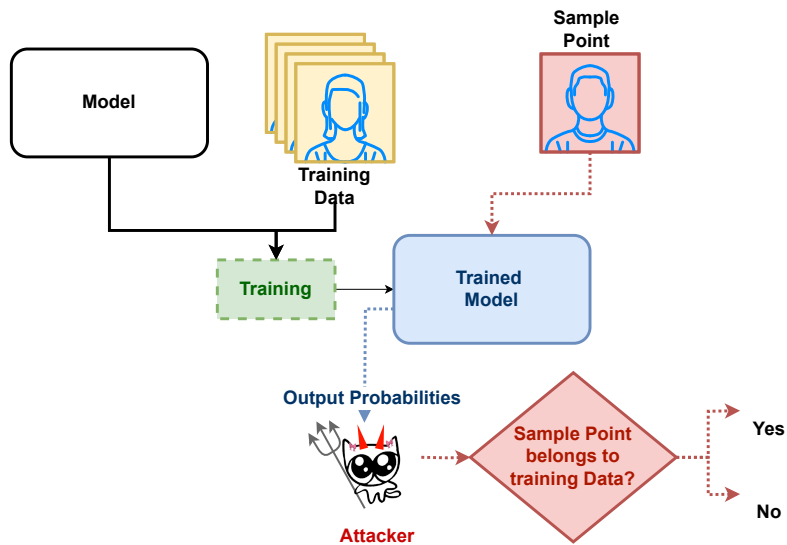


Figure 4 Illustration of the general Membership Inference Attack (MIA) framework. An attacker attempts to determine whether a specific input was part of the target model's training dataset based on its predictions.

Introduced by Shokri et al., 2017, These are techniques where given an input point, and access to a trained predictive model, an attacker¹ tries to guess if the input point was a part of the dataset used to train the predictive model. MIA can be broadly classified into two kinds:

2.3.1 White-box Membership Inference Attacks

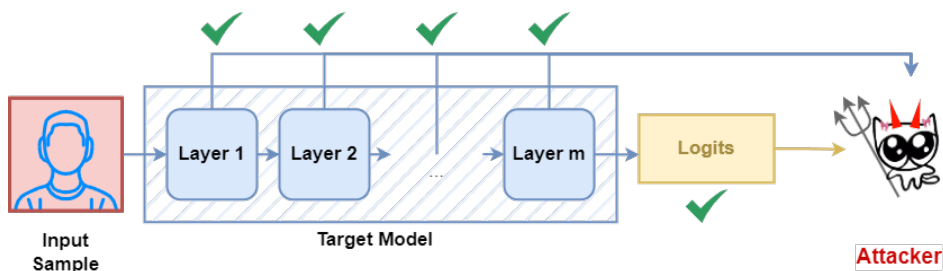


Figure 5 Visualization of the White-box Membership Inference Attack (MIA) setup. In this scenario, the attacker has full access to the model's intermediate layers.

In White-box MIA's, the attacker has access to all the information about the target model, such as the model architecture, learned parameters, training configuration, and possibly even the distribution of the training data.

¹ Attacker graphic by Jelena, <https://es.pinterest.com/JelenaOFC/>

2.3.2 Black-box Membership Inference Attacks

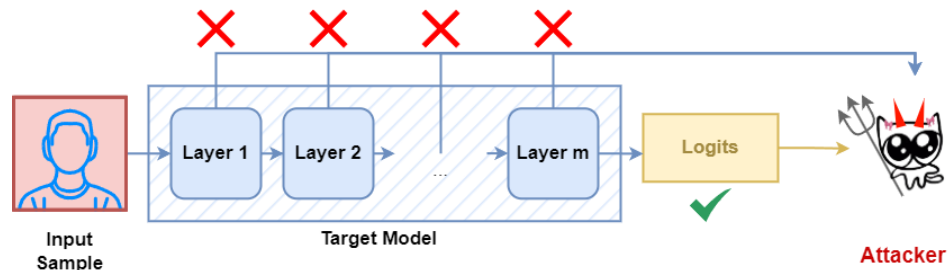


Figure 6 Visualization of the Black-box Membership Inference Attack (MIA) setup. In this scenario, the attacker only has access to the model's prediction outputs.

In black-box MIA's, the attacker has limited access to the target model's training data distribution and only gets the prediction outputs of the target model. In this paper, we've only run tests for the Black-box settings since mostly in practice, organizations releasing their trained models to the public, provide black-box access to their models through their APIs.

3 Methodology

3.1 Masked Autoencoder

This work extends the MAE framework (Figure 3), which is a powerful Autoencoder (Ballard, 1987; Rumelhart et al., 1986) based MIM technique, built on the ViT architecture. The main elements of MAE are:

3.1.1 Masking

MAE works by first splitting the images into patches of equal size. A random subset of these patches are selected for the MAE encoder. The rest are referred to as masked patches and are kept as target for the decoder to predict. The original MAE paper suggests using a high masking ratio (about 75%) for the encoder to learn meaningful representations and avoid shortcuts like extrapolating missing pixels based on neighbouring patches.

3.1.2 Encoder

The encoder in the MAE architecture follows the standard ViT setup. The exact model configuration of the encoder in our experiments are written in the following Results section. High masking ratios for the patches reduces the computation cost of the encoder, since it does not have to operate on the masked patches.

3.1.3 Decoder

The decoder takes the encoder's output for the unmasked patches, and also learnable mask tokens for the missing masked patches. Unlike Masked Language Modeling (MLM), where masked tokens are added before passing the input to the first transformer block, here the masked tokens are added after computing the encoder's output and are then passed to the decoder to make it computationally less expensive. The decoder then produces an output which is a reconstruction of the original masked image but with the missing positions filled in. MAE uses a lightweight decoder since the quality of the reconstructed images is not as important to us as the underlying latent representation.

3.1.4 Reconstruction Loss

MAE tries to reconstruct the masked pixels directly, so the reconstruction loss is a simple Mean Squared Error, applied to the pixels directly. The loss is formally defined as follows:

Let $\mathbf{x} \in \mathbb{R}^{3HW \times 1}$ be the input pixel RGB values and $\hat{\mathbf{x}} \in \mathbb{R}^{3HW \times 1}$ be the predicted pixel values, where H denotes the height of the image and W denotes the width of the image. The MAE Reconstruction loss can then be written as

$$L_{MAE} = \frac{1}{\Omega(\mathbf{x}_M)} \sum_{i \in M} (\hat{\mathbf{x}}_i - \mathbf{x}_i)^2 \quad (3.1)$$

where M denotes the set of masked pixels, $\Omega(\cdot)$ is the number of elements, and i is the pixel index.

3.2 Knowledge Distillation

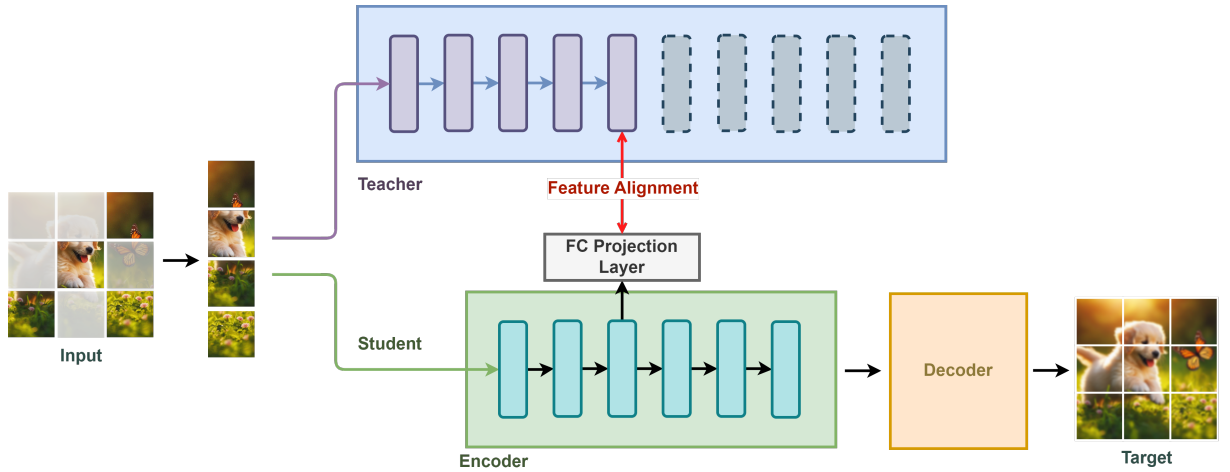


Figure 7 Illustration of the Distillation Framework for MAE. This framework works by distilling intermediate layer features from a large teacher model to a smaller student model.

The DMAE framework seeks to combine Knowledge Distillation with MAE to achieve small and fast models with performance comparable to powerful but slower large models. The original paper empirically finds that currently existing KD techniques for ViTs such as DeiT (Touvron et al., 2021) do not work well for MAE, so the authors take an alternative approach of applying KD directly in the pre-training stage.

Due to absence of categorical labels for pre-training MAE, response-based distillation is not very straightforward, so DMAE resorts to distilling the intermediate layers.

First the intermediate feature output of a specified layer of the student model is taken and fed into a fully-connected projection head. The projection head also addresses the problem of possible feature dimension mismatch between teacher models and student models.

Let $\mathbf{z}_k^S \in \mathbb{R}^{LC_S \times 1}$, $\mathbf{z}_l^T \in \mathbb{R}^{LC_T \times 1}$ be the features extracted from the k th layer of the student model and the l th layer of the teacher model, respectively, where L denotes the patch numbers, and C_S, C_T denote the channel dimension of the student and the teacher respectively. $\sigma(\cdot)$ denotes the projection network function. The feature distillation loss L_{Dist} can be written as:

$$L_{Dist} = \frac{1}{\Omega(\mathbf{z}_l^T)} \sum_i \|\sigma(\mathbf{z}_k^S)_i - \mathbf{z}_{l,i}^T\|_2. \quad (3.2)$$

The final loss for pre-training can then be expressed as the MAE (eq. 3.1) reconstruction loss and the distillation loss (eq 3.2), weighted by a hyperparameter α :

$$L = L_{MAE} + \alpha \cdot L_{Dist} \quad (3.3)$$

3.3 Membership Inference Attack

The original method proposed by Shokri et al., 2017, works by training multiple "shadow models". Each shadow model has the same architecture as the target model but is trained on a separate, independent but identically distributed subset of the data. Then for each shadow model, it's outputs are calculated for the samples in the training dataset (member samples) and samples which were not in it's training dataset (non-member samples). An attacker model is then trained on these outputs, which it classifies as coming from a member/non-member. However, training multiple shadow models is computationally expensive and time consuming. Future works on MIA have produced computationally less expensive methods. We use two such methods, as follows:

3.3.1 Single Shadow Model based Attack

A variant of shadow model attack, introduced by Salem et al., 2019, shows that it's sufficient to train a single shadow model. This variant achieves performance similar to Shokri et al., 2017 while significantly reducing the computational cost.

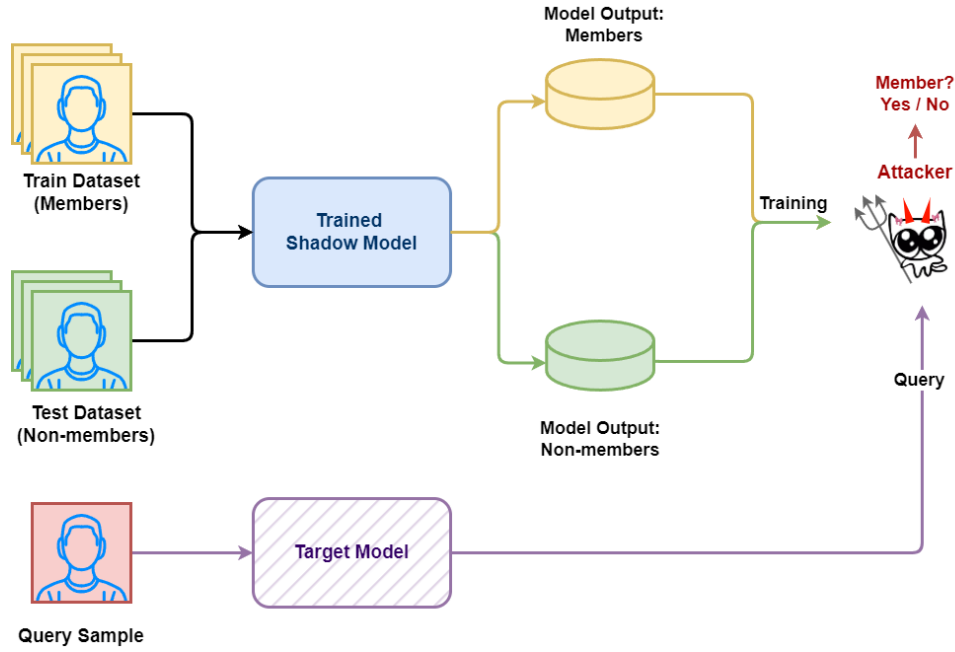


Figure 8 Illustration of the Single Shadow Model Based Membership Inference Attack Framework. This approach utilizes a single shadow model to approximate the target model’s behavior.

For our experiments, we use a simple multilayer perceptron as the attacker model.

3.3.2 Threshold based Attack

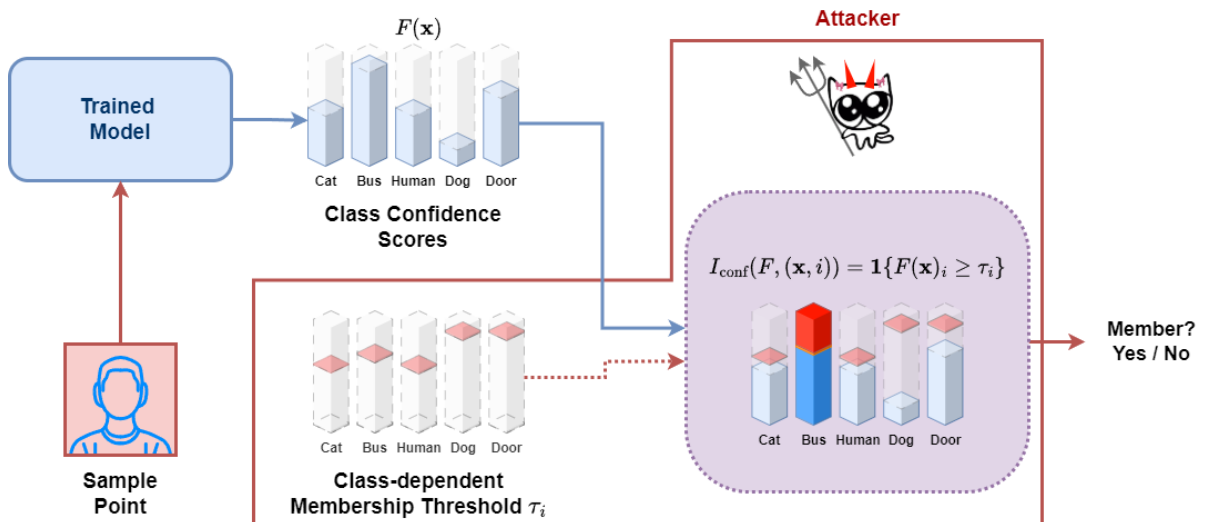


Figure 9 Illustration of the Threshold Based Membership Inference Attack Framework. This approach estimates individual thresholds on prediction confidence for each class to determine membership status, classifying an input as a member if its confidence exceeds the estimated threshold.

MIA based on using a threshold on prediction confidence (Song et al., 2019; Yeom et al., 2018) achieve similar accuracy as Neural Network based attacks. This is based on the intuition that while training, the target model minimizes prediction loss over training data, which means that the prediction confidence of a training sample should be close to 1. While on the contrary, the model would usually be less confident while predicting for a test sample. Thus the attacker can then classify an input example as a member if it's prediction confidence exceeds a certain threshold. We formally define this as:

$$I_{\text{conf}}(F, \mathbf{x}, i) = \mathbb{1} \{F(\mathbf{x})_i \geq \tau_i\} \quad (3.4)$$

where τ_i is a threshold for each class label i (Song and Mittal, 2021).

4 Experiments and Results

4.1 Model Configuration

Model	Embedding Dimensions	Depth	Attention Heads
Teacher (ViT-Tiny)	192	12	3
Student Encoder	64	6	4
Student Decoder	48	2	4

Table 1 Teacher and Student Model Configurations

The original DMAE paper discusses distilling from ViT-L to ViT-B, whereas our approach distills from ViT-Tiny to even smaller ViTs for pre-training. The specifics of the configurations are given in Table 1. Since our study is not focused on accuracy or model size, but rather on investigating resistance against Membership Inference Attacks, it is sufficient to use smaller models.

For pre-training, we train both the teacher and student models using the Adam optimizer with a base learning rate of 0.001, optimizer momentum $\beta_1 = 0.9$ and $\beta_2 = 0.999$. We pre-train both the models on CIFAR-100 for 75 epochs, with the teacher model achieving a test accuracy of 52.06%. Throughout all our experiments, both for pre-training and downstream classifications, the original image sizes were 32x32 RGB, scaled up to 72x72. The images were then divided into patches of size 6x6. We also use a masking ratio of 75% as suggested in the original MAE paper.

For layer alignment, as suggested in the original DMAE paper, we align the roughly middle layers of the teacher and student models. Specifically, we align the output of the 4th layer of the student encoder with the output of the 5th layer of the teacher model. Both the ViT with feature distillation and the vanilla ViT share the exact same architecture as our MAE encoder.

Pre-training Technique	Learning Rate
DMAE	0.001
ViT with feature distillation	0.001
MAE	0.0001
ViT (Vanilla) - no pre-training	0.0001

Table 2 Learning rates for different pre-training techniques during downstream classification.

While training the downstream classifiers on CIFAR-10, we use the Adam optimizer with the same hyperparameters as in the teacher training, except for the learning rate,

Model	Mean Accuracy (%)
MAE	71.65
ViT with Feature Distillation	72.73
ViT (Vanilla) - no pre-training	72.70
DMAE	76.45
DMAE (Min Val Loss)	73.47

Table 3 Mean Accuracy of Different Models

which varies depending on the pre-training technique (as shown in Table 2). We found that going above or below these specific learning rates led to model divergence or excessively slow training.

4.2 Model Training

We create two checkpoints for DMAE: one based on the minimum validation loss and another based on the maximum validation accuracy. While MAE speeds up training, in our experiments, we find it's performance to not be significantly different from the ViT model without pre-training. However, the variant with Knowledge Distillation but without MAE training consistently achieved higher validation accuracy over time.

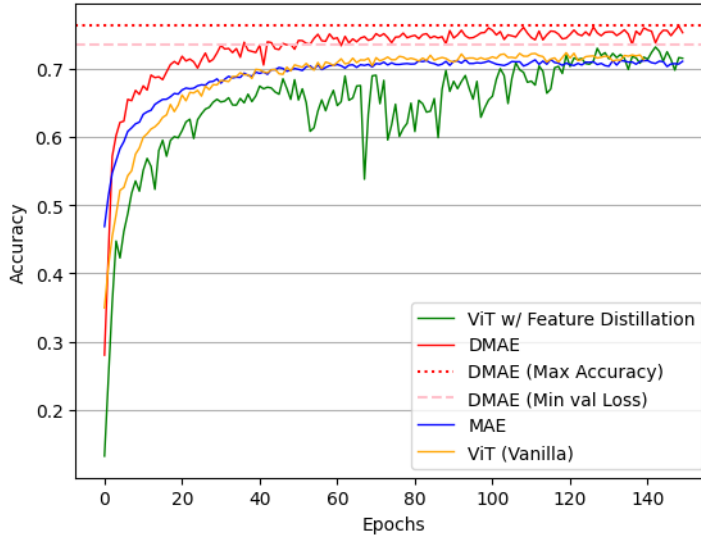


Figure 10 Validation Accuracy over Epochs for Downstream Classification.

DMAE not only trains faster but also achieves competitive accuracy as shown in Table 3. When trained for more epochs, it reaches even higher accuracy. Notably, at the checkpoint corresponding to minimum validation loss, DMAE already reaches higher accuracy than that of all other models (pink dashed line in Figure 10).

4.3 Membership Inference Attack

4.3.1 Attacker’s Performance Measure

We use the Receiver Operating Characteristic (ROC) curve and its associated Area Under the Curve (AUC), to measure the attacker’s performance. The ROC curve plots the true positive rate (i.e., correctly classified member samples) against the false positive rate (i.e., non-member samples incorrectly classified as members) by varying the classification thresholds.

The AUC score ranges from 0.5 (random guess) to 1 (perfect classifier), this is used as a metric for measuring how vulnerable a model is to MIA. A higher AUC indicates that the attacker is better at distinguishing between member and non-member samples, suggesting greater susceptibility to membership inference. While on the other hand, an AUC score closer to 0.5 indicates that the model is more resilient to such attacks, as the attacker struggles to differentiate between the two types and is no better than a random guesser. To learn more about ROC curves, please refer to Fawcett, 2006.

4.3.2 Results

In Table 4 and Table 5, we compare the results of performing MIA across various pre-training techniques, using the CIFAR-10 dataset for evaluation. We also compare them visually in Figure 11 and Figure 12

Method	Data Subset	
	Incorrectly Classified	Entire Dataset
MAE	0.6186	0.5469
ViT w/ Feature Distillation	0.5802	0.5519
Vanilla ViT (no pre-training)	0.6213	0.5634
DMAE	0.6731	0.5678
DMAE (Min Val Loss)	0.5227	0.5315

Table 4 MLP based attack : Average attacker’s AUC Score (lower is better)

Samples which were incorrectly classified by the model, were found to be the most vulnerable against MIA. We also compared performance on the entire dataset alongside the incorrectly classified subset.

A surprising outcome of the experiments is that DMAE, when trained until it reached the minimum validation loss (achieving the second highest accuracy), demonstrates

Method	Data Subset	
	Incorrectly Classified	Entire Dataset
MAE	0.6431	0.5657
ViT w/ Feature Distillation	0.5989	0.5451
Vanilla ViT (no pre-training)	0.6387	0.5639
DMAE	0.6929	0.5757
DMAE (Min Val Loss)	0.5570	0.5318

Table 5 Threshold based attack : Average attacker’s AUC Score (lower is better)

the greatest resistance to MIA. However, when trained for a greater number of epochs, DMAE achieved the highest accuracy but also becomes the most vulnerable to MIA.

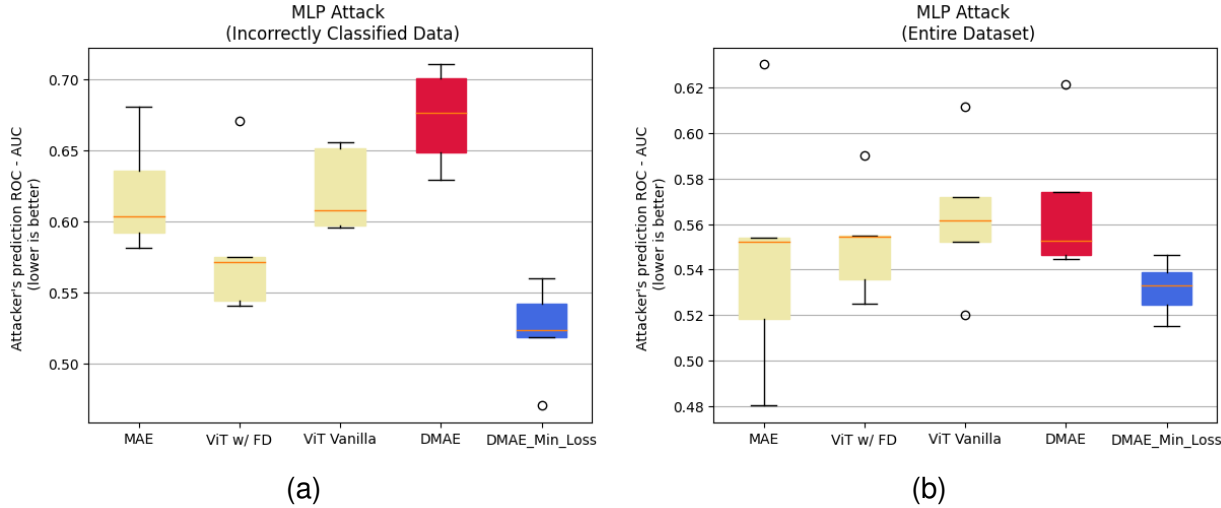


Figure 11 Attacker’s Prediction AUC Score for MLP based attack performed on the incorrectly classified subset (a), and the entire dataset (b)

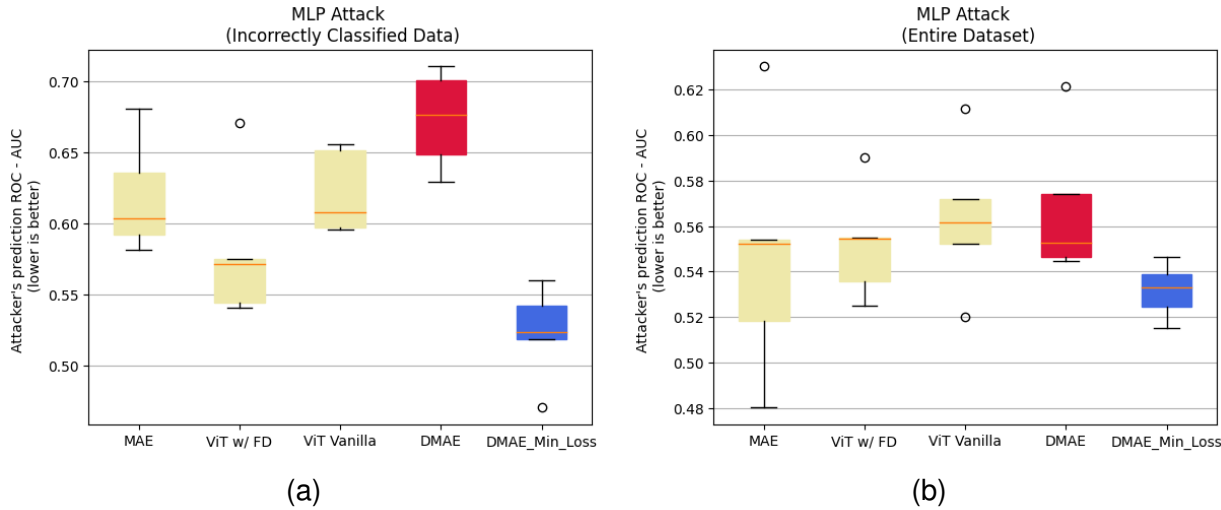


Figure 12 Attacker’s Prediction AUC Score for Threshold based attack performed on the incorrectly classified subset (a), and the entire dataset (b)

For the incorrectly classified subset, DMAE achieved an average AUC of 0.6731 for the MLP attack and 0.6929 for the threshold attack, while DMAE (Min Val Loss) had an average AUC of 0.5227 for the MLP attack and 0.5570 for the threshold attack.

4.3.3 Relation to Training and Validation Loss

The reason for the significant increase in vulnerability to MIA of DMAE can be attributed to overfitting. Shokri et al., 2017; Truex et al., 2019; Yeom et al., 2017; Yeom et al., 2018 show that the degree of over-fitting of the target directly influences its vulnerability to MIA.

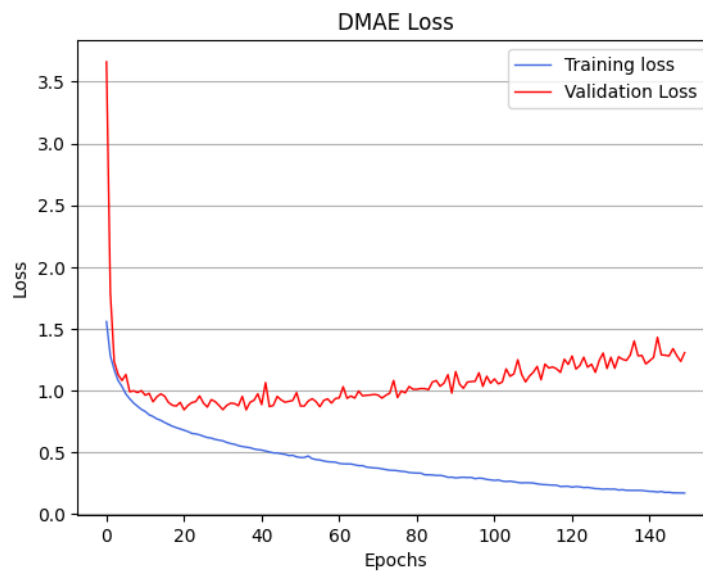


Figure 13 Comparison of Training and Validation Loss during Downstream Classification over Epochs. The training loss continues to decrease while the validation loss increases after a certain point, indicating overfitting.

By comparing the validation loss and training loss over epochs (Figure 13), we see that after a certain point, the validation loss starts to increase, while the training loss still keeps decreasing. However as seen in Figure 10, the validation accuracy keeps increasing regardless. As explained by Guo et al., 2017, this can be attributed to the fact that accuracy is only dependent on the majority class in the output predictions, while cross-entropy loss is also dependent on the confidence of the predictions. While overfitting, the confidence output for the majority class for the validation dataset keeps decreasing, but on the other hand, its ability to predict the correct class as the majority class keeps improving. Meanwhile the confidence for the majority class keeps increasing for the training data. Vulnerability to MIA increases with difference in prediction confidence between the training and the validation datasets, resulting in the effect seen in section 4.3.2.

5 Conclusion

In this paper, we explored the use of Masked Autoencoder with Knowledge Distillation for pre-training Vision Transformer based models to have improved privacy. Our results demonstrate that this method not only improves the model's performance but also exhibits more resistance to Membership Inference Attack.

However we also observe that with extended training with this method, improves accuracy significantly, but this comes at a cost of increased vulnerability to Membership Inference Attack. This highlights the inherent privacy-utility trade-off of such models.

In section 4.3.3, we analyze this behaviour and attribute this to over-fitting. We also explain how such models show gradual improvement in validation accuracy while the validation loss keeps increasing.

5.1 Future Scope

Yang et al., 2023 proposed Privacy-Preserving Generative Framework (PPGF), for improving a model's privacy. This method first fits a Variational Autoencoder (VAE) (Kingma and Welling, 2014) to the original dataset. A synthetic dataset is generated by sampling latent codes from the encoder's output distribution and passing it to the decoder. A classification model is then trained on the VAE generated synthetic data. The PPGF paper shows that this helps train models resilient to MIA while also maintaining high utility.

The DMAE framework could be extended to use techniques similar to PPGF to address the privacy degradation problem in section 4.3.2. Moreover, since the downstream classification for DMAE works with the latent codes of the encoder network, sampling from the encoder's output distribution, should suffice for synthetic data generation. This would save additional computational cost which is otherwise incurred by passing the latent codes to the decoder.

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Fürth, 16th October 2024

A handwritten signature in black ink, appearing to read 'Safalya Pal', written in a cursive style.

Safalya Pal